TOUCH SCREEN BASED AUTOMATION SYSTEM

Submitted in partial fulfillment of the requirements

of the degree of

Bachelor of Engineering

by

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This is to certify that the project entitled **"Touch Screen Based Automation System"** is a bonafide work of the following students

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Project Report Approval

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Examiners

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2.-- Date:

Place:

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ACKNOWLEDGMENTS

Special thanks to our Guide ***Prof. Shruti Patil*** for assisting us to partially complete our project on ***"Touch Screen based Automation System".***

Her expertise and talent in circuit designing and troubleshooting and logical regression helped us effectively to partially complete this project.

We would also like to thank our Head of Department ***Mrs. Swapna Patil*** madam for providing us facility and labs, which helped us constantly in increasing our technical knowledge, and to write this report.

We are also thankful to our Principal ***Dr. Deven Shah*** sir for his continuous encouragement throughout the process.

Now, last but not the least special thanks to all the staff of **Electronics & Telecommunication Engineering Department** for their technical support and constant motivation, without which this work would not have become successful.

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ABSTRACT

TOUCH SCREEN BASED AUTOMATION SYSTEM

Home automation refers to the use of computer and information technology to control home appliances and features. Systems can range from simple remote control of lighting through to complex computer/micro - controller based networks with varying degrees of intelligence and automation. Home automation is adopted for reasons of ease, security and energy efficiency.

As the number of controllable devices in the home rises, interconnection and communication becomes a useful and desirable feature. .In simple installations, automation may be as straightforward as turning on the lights when a person enters the room. In advanced installations, rooms can sense not only the presence of a person inside but know who that person is and perhaps set appropriate lighting, temperature, music levels or television channels, taking into account the day of the week, the time of day, and other factors.

Home automation can also provide a remote interface to home appliances or the automation system itself, to provide control and monitoring on a smartphone or web browser.

A touch panel is interfaced to the microcontroller on transmitter side which sends ON/OFF commands to the receiver where loads are connected. By touching the specified portion on the touch screen panel, the loads can be turned ON/OFF remotely through wireless technology. The loads are interfaced to the microcontroller.

1.INTRODUCTION

1.1. **Motivation and Background**

As technology is advancing so houses are also getting smarter. Modern houses are gradually shifting from conventional switches to centralized control system, involving touch screen switches. Presently, conventional wall switches located in different parts of the house makes it difficult for the user to go near them to operate. Even more it becomes more difficult for the elderly or physically handicapped people to do so. Remote controlled home automation system provides a simpler solution with touch screen technology.

1.2. **Objective**

Our objective is to design a home automation system which uses a touch screen to control the various devices of the house like light, fans ,TV etc. The communication range is also long and there is no loss of data.

1.3. **Outline of the Dissertation**

Traditional home control systems require manual operation of individual devices, leading to inconvenience, inefficient energy use, and limited accessibility. This project aims to develop an integrated, user-friendly home automation solution that allows centralized, remote control of multiple household devices through various interfaces, improving convenience, energy efficiency, and accessibility for homeowners.

2.Problem Statement

The need for efficient home management has grown, as traditional appliance controls are cumbersome and inefficient. This project uses an Arduino Mega to develop a touchscreen home automation system, allowing users to control appliances via a smartphone app, touch sensor, and IR remote. The system provides a unified platform for simplified control, enhancing accessibility and energy efficiency.

3. Literature Survey

Year of publication	Author	Publication Paper	Application
Feb-21	Samruddhi Fasi, Neha Khanaj, Pratiksha Lokare, Poorva Shreshthi, Ms.R.R.Gaji	Home automation system	The GSM-based system enables remote control of home appliances via smartphone, enhancing convenience and security.
Apr-19	Kondapalli Jayaram Kumar, Chodisetti Venkata Balaji, Sukkireddy V V Manikanta Swamy Naidu, Vendra Vamsidhar, Desaneedi Satish	CAPSENSE BASED TOUCH PANEL CONTROLLED HOME AUTOMATION	The Capsense-based system enables touch and app-based control of home appliances via loT, providing convenience and accessibility for all users.
Nov-18	Prof. Ami Munshi, Roshan Gosalia(Student), Darshil Gala(student)	Android Based Home Automation System	This application demonstrates the integration of mobile technology with home electronics, providing a step towards creating a more connected and efficient living environment.
Mar-18	Prof. Manohar Wagh, Vrushabh Gadhari, Harshad Sonawane, Shriram Shekar, Rahul Mahala.	Touch screen based automation system	This system aims to provide a user-friendly, secure, and efficient way to manage various aspects of a home environment through a centralized touch screen interface.
Jun-14	Mr.Yash Inaniya, Naresh Kumari and Urvashi Luthra.	Automation System using capacitive screen	The paper emphasizes that this system is designed to be cost-effective, using readily available materials, and can be expanded to control more devices beyond the four demonstrated in the project.

LITERATURE SURVEY HAS BEEN IMPLEMENTED:

3.1. Samruddhi Fasi , Neha Khanaj, Pratiksha Lokare, Poorva Shreshthi, Ms.R.R.Gaji, "Home automation system", February 2021, at Sharad Institute of Technology Polytechnic.

Abstract:

Electronic devices and appliances have become very common in this recent year of technology especially with fast development in smartphones. In this paper, the design of Home Automation System compatibly with Local housing and good features for home automation via remote access are presented. GSM Based Home Automation System Using Android and Arduino is design and implemented. In this research work a part of smart home technology which using GSM in a mobile device is used, so it will cheap and efficient to use. This paper describes about home automation system which would use to enable home lighting, garage door motor, water pumping motor and smoke detection using a smart phone application with GSM wireless technology

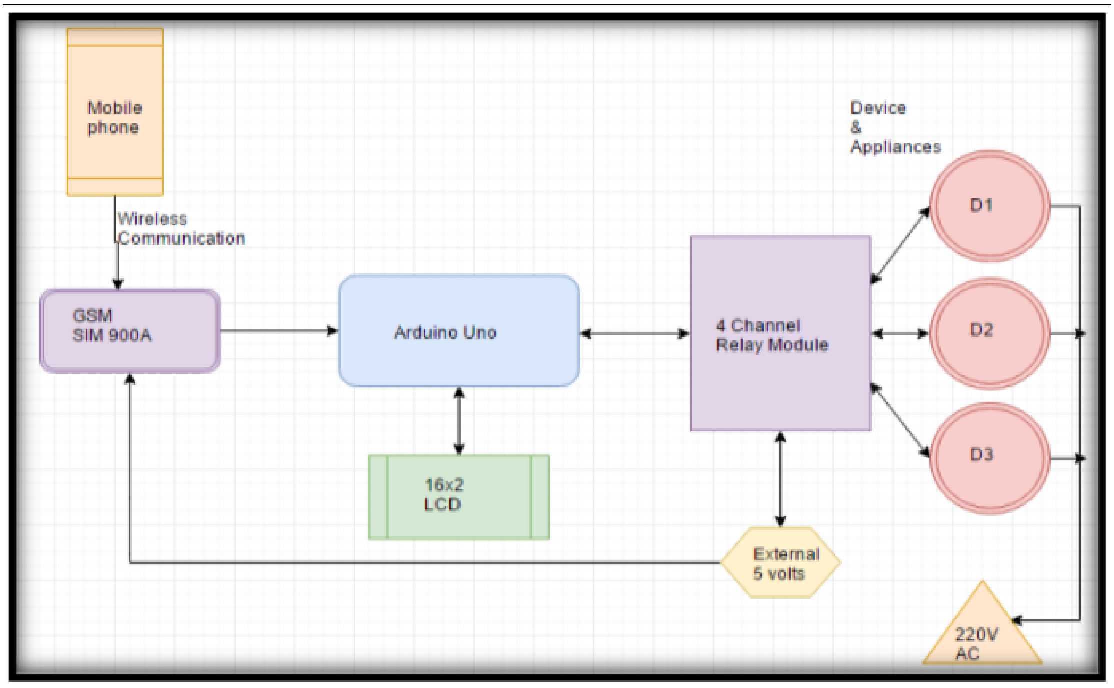

Figure 3.1.1

GPS module that has been set to slave mode is used to communicate the controller with the smart phone application. Application is created by App Inventor 2. App Inventor 2 is a visual, blocks language for building Android Apps. There are two main types of components in an app

Inventor 2, visible and nonvisible. Visible components of application are can see when the app is launched that includes buttons, text boxes, and labels. These are often referred to as the graphical user interface (GUI). Non-visible components are not seeing, so they are not part of the user interface. Instead, they provide access to the built-in functionality of the device. The non-visible components are the technology within the device they are little worker bees and they do jobs for control of the application. App Inventor 2 can easily create GUI interface for user friendly and block editor can make the relevant function of each button from application easily without writing coding.

3.2. Kondapalli Jayaram Kumar, Chodisetti Venkata Balaji, Sukkireddy V V Manikanta Swamy Naidu, Vendra Vamsidhar, Desaneedi Satish, "CAPSENSE BASED TOUCH PANEL CONTROLLED HOME AUTOMATION", at Godavari Institute of Engineering and Technology

Introduction:

Today in the headway of Automation innovation, life is getting simpler and less demanding in all spheres. Home automation is a modern technology that modifies your home to perform different sets of task automatically. Today Automatic frameworks are being favored over manual frameworks. No wonders, home automation in India is already the buzz word, especially as the wave of second generation home owners grows, they want more than shelter, water, and electricity. The first and most obvious advantage of Smart Homes is comfort and convenience, as more gadgets can deal with more operations (lighting, temperature, and so on) which in turn frees up the resident to perform other tasks. Smart homes filled with connected products are loaded with possibilities to make our lives easier, more convenient, and more comfortable. The requirement for Office and Home automation arises due to the advent of IoT, in a big way in homes and office space. The smart home/office gadgets interact, seamlessly and securely control, monitor and improve accessibility, from anywhere across the globe. These smart automation devices happen to have an interface with IoT. IoT automation will be the key to bridging the gap between human limitations and technology's capabilities. The IoT based Home Automation will enable the user to use a Home Automation System based on Internet of Things (IoT). The modern homes are automated through the internet and the home appliances are controlled. The user commands over the internet will be obtained by the Wi-Fi modems. The Microcontroller has an interface with this modem. This is a typical IoT based Home

Automation system, for controlling all your home appliances. The beauty of the Home Automation system lies in the fact that the settings are manageable from your smart phones and other remote-control devices. Smart home IoT devices can help reduce costs and conserve energy. The Home Automation segment includes smart lighting, smart TVs and other appliances. Wearable's (Smart Watch, fitness brands, smart headphones, smart clothing) are also expected to witness the growth in the future. IoT is really the secret that makes this whole system work. Today in India, nearly 22.5 per cent of the consumers surveyed were familiar with the concept of IoT, with maximum awareness seen in the 36-55 age groups which clearly indicates that there is immense opportunity for increased adoption of such technologies.

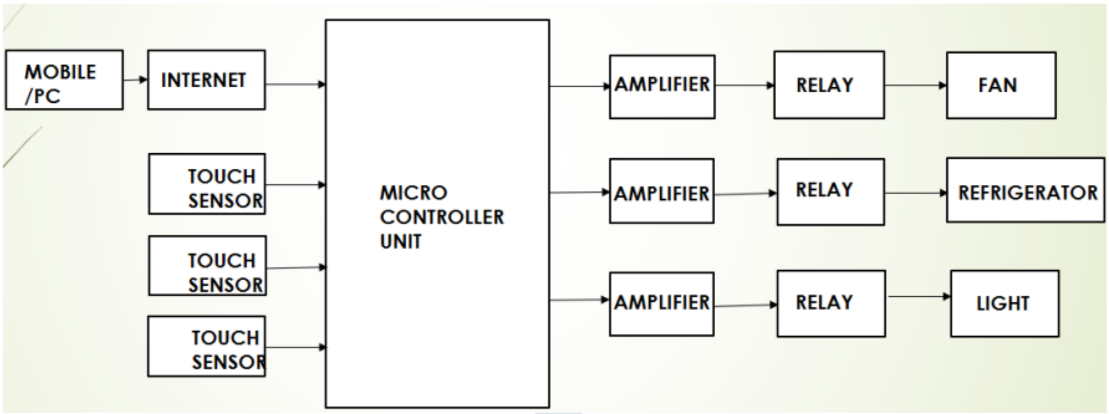

Figure 3.2.1

A. TOUCH SENSOR: When there is contact with the surface of the capacitive touch sensor, then there exist some change in capacitance in sensor and the circuit is closed inside the sensor and there is flow of current. The measurement circuit will detect the change in capacitance and convert it into a triggering signal

B. NodeMCU is an open source IoT platform. It includes firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. The term "NodeMCU" by default refers to the firmware rather than the development kits. Memory: 128kBytes Operating system: XTOS Developer: ESP8266 Open source Community Storage: 4Mbytes

C. A relay is an electromagnetic switch operated by a relatively small electric current that can turn on or off a much larger electric current. The heart of a relay is an electromagnet (a coil of wire that becomes a temporary magnet when electricity flows through it). The Coil Terminals control the switch. When voltage is applied across the coil it becomes an electromagnet. Its core attracts the switch armature and activates the switch. And diverts power supply to the home appliance connected at end terminal of relay.

D. Node MCU is IoT ready module and it works on 3.3V only but controlling relay works on 5V so it was a challenges. In this project, 3.3V to 5V converter is designed. Here NodeMCU 5V supply is used to power 3.3-5V Converter from Vin pin and 3.3V Supply is used from 3.3V Pin. All GPIO Pins of NodeMCU provide 3.3V logic whenever it is high so it is provided to trigger the BC547 transistor

3.3. Prof. Ami Munshi, Roshan Gosalia, Darshil Gala, "android based automation system", November 2018, at International Journal of Scientific & Engineering Research Volume 9, Issue 11

Introduction:

In the present scenario the majority of switching operations are manual and do not imbibe the idea of IOT and the interconnection of various applications to help optimize operation. These days there is a clear divide between electrical and software systems, and this leads to inefficient and often incompatible processes. To solve this dearth in integration of a variety of applications, this project of Home Automation aims to use a modem that brings all switches and control to the user in one place.

Overall Schematic:

As far as automation goes, ergonomics is the essence of all function. Therefore in our project we implement the usage of a modem that follows such a philosophy.

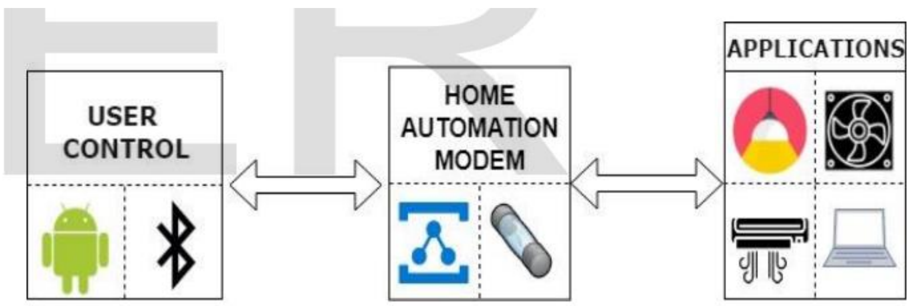

Figure 3.3.1

The above diagram shows the overall schematic flow that illustrates user control, the modem's functions and the end applications to be operated. First we see that the user operates an android based smartphone on which an application geared towards automated control is installed. This

Android application includes support for the Bluetooth network stack through which it allows data transfer and control of another Bluetooth device and its parameters. Then we see the heart of the project, the Automation Modem wherein a combination of a microcontroller, fuse and a relay system form the processor that takes in input data such as Bluetooth packets and voice data and then sends it to a rely system that is responsible for interfacing with applications. This relay system uses a relay driver that drives four relays used to drive four external applications.

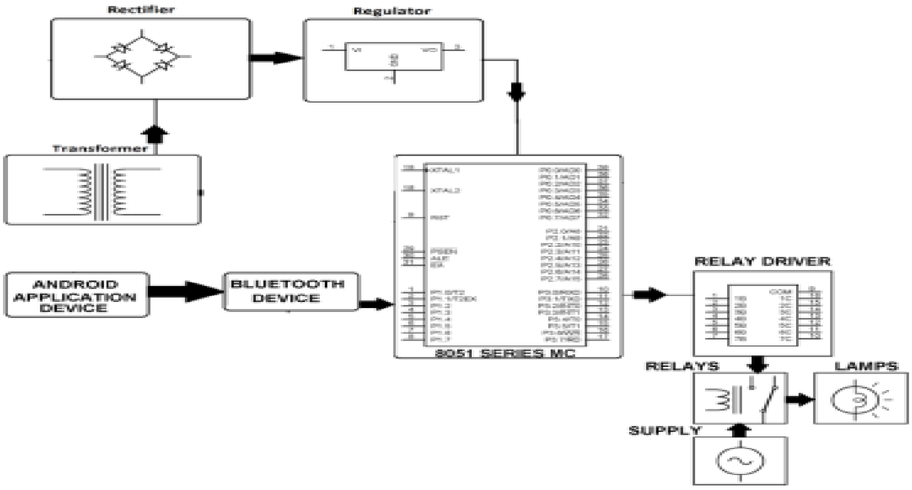

Figure 3.2.2

Implementation:

We have implemented a minimal effort and proficient shrewd home framework in our outline. This framework has two principle modules: the hardware interface module and the software communication module. At the core of this framework is the AT89s52 microcontroller which is additionally equipped for working as an interface for all the hardware modules. All communication and controls in this framework go through the microcontroller. The shrewd home framework offers changing functionalities to control lighting, fans/aeration and cooling systems, and other home machines associated with the transfer framework. All these can be controlled from the Android advanced mobile phone application.

Smart Home Systems give interface between different kinds of home and electrical devices like bulbs, TVs, fans, curtains and so forth. This system gives control and convenience of the devices to the client according to client's requirements. Subsequent to breaking down other existing frameworks, we propose the novel system for better human association and for giving better usage of android and arduino. By utilizing home computerization framework we can oversee cost, adaptable and vitality proficient savvy homes. The home computerization framework has been tentatively demonstrated to work palatably by associating test machines to it and the devices were effectively controlled from a remote cell phone. The Bluetooth customer was effectively tried on a huge number of various cell phones from various producers, accordingly demonstrating its versatility and wide similarity. This undertaking won't just give accommodation to the normal man yet will be a help for the elderly and debilitated.

3.4. Prof.Manohar Wagh, Vrushabh Gadhari, Harshad Sonawane, Shriram Shelar ,Rahul Mahale "touch screen based automation system", March 2018,At International Research Journal of Engineering and Technology (IRJET)

Abstract:

- In recent years, the home environment has project focuses on assisting the users to control as well as to know the exact status of electric appliances in their home at that instant by using GSM and Zig-Bee which is wireless communication. Previously home automation are very complicated based on hardware. Thus it is difficult to maintain Factors like security, reliability, usefulness, robustness and price. Now a days it consist of touchscreen which easy to use. Now that human and computer interaction has been developed into a more wide and sophisticated field., designing and operating of intelligence system has been more user friendly than ever. Home automation is a system that helps a user to operate switching various appliances and lighting devices from a single input. The touch screen used as input is much simpler to operate. Touch screen has been widely accepted as the most comfortable input to be provided to the user. Not only they are easy to operate but they also give a sense of personal involvement which the user always appreciate.

Introduction:

Now a days, as rapid growth of technologies to reduce the human efforts. Home automation is one of the technology which is used to controlling as well as monitoring household appliances. There are different technologies to controlling by home automation using Wi-Fi, Zigbee, Android, Raspberry pi etc.As Technology is advancing so houses getting smarter. In our system ,as in automation GSM used for reducing complexity of automation.GSM and Zig-Bee used as wireless communication in Homes to control all Home appliances in home.control access used command for better access[2].Next paper is on the home automation using Zigbee, in that they mainly focused on:

1. Wi-fi based home automation: The component of the system will always be connected.

A. Wi-Fi:

1. Each User must have a User ID and password

2. There is only one Administrator. Server must always run under windows system. There should be Internet connection available.

3. Efficient use of Hardware and Software is user friendly.

B. Bluetooth: Low cost but range problem so they use ARM 9.

C. Zigbee: This system has attractive features such as SMS-Email notifications. In this perspective, ZigBee is emerging network technology as a wireless communication standard that is capable to satisfy such requirements

System Architecture:

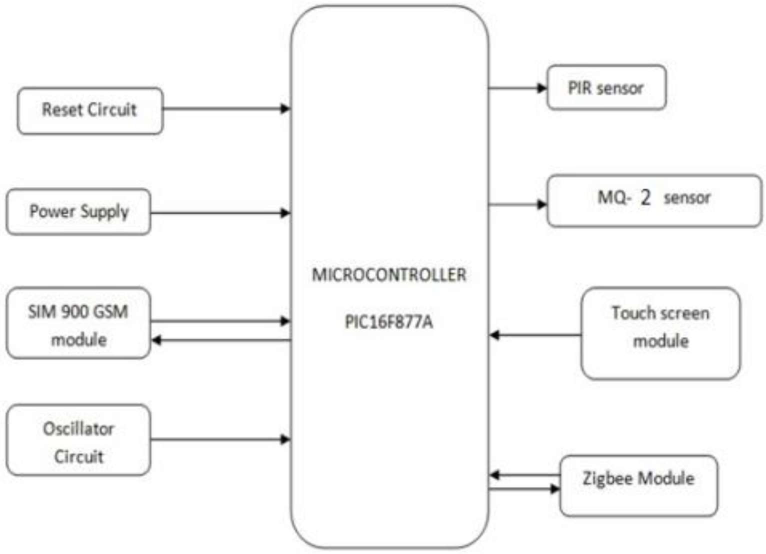

Figure 3.4.1

Transmitter Algorithm

1. Start

2. Initialize 162 Alphanumeric LCD, Zigbee S2

3. Initialize ADC and UART registers of PIC16F877A

4. Read the ports where touch sensor is interfaced

5. If (Touch key 1 == `1') Then transmit data `1' serially through Zigbee Else If (Touch key 2 == `1') Then transmit data `2' serially through Zigbee Else If (Touch key 3 == `1') Then transmit data `3' serially through Zigbee Else If (Touch key 4 == `1') Then transmit data `4' serially through Zigbee Else If (PIR == 1) Send message Else If (LPG > = 150)

6. Go to step 4

7. Stop

Receiver Algorithm:

1. Start
2. Initialize Zigbee S2 module and ports where relay is interfaced
3. Initialize UART registers of PIC16F877A
4. Receive byte serially through Zigbee S2
5. If (Received byte == ` 1 ') Then drive device 1 through relay Else If (Received byte == ` 2 ') Then drive device 2 through relay Else If (Received byte == ` 3 ') Then drive device 3 through relay Else If (Received byte == ` 4 ') Then drive device 4 through relay
6. Go to step 4
7. Stop

3.5. Mr.Yash Inaniya, Naresh Kumari and Urvashi Luthra "Home automation using capacitive touchscreen", June 2014, al Int. Journal of Engineering Research and Applications ISSN : 2248-9622, Vol. 4, Issue 6(Version 2)

Abstract:

Technology has been constantly evolving and with the advent of touchscreen in human life , devices are much easier and simple to operate. This work is mainly focused on building home automation system which is more user friendly and thus can be operated by anyone. Earlier home automation systems were completely mechanically operated and thus required a lot of maintenance and were costly also. Now that human and computer interaction has been developed into a more wide and sophisticated field , designing and operating of intelligence system has been more user friendly than ever. Home automation is a system that helps a user to operate switching various appliances and lighting devices from a single input. The touchscreen used as input is much simpler to operate. Touchscreen has been widely accepted as the most comfortable input to be provided to the user. Not only they are easy to operate but they also give a sense of personal involvement which the user always appreciate. The materials used in this system are easily available in the local market so that the touch screen sytem is cost effective

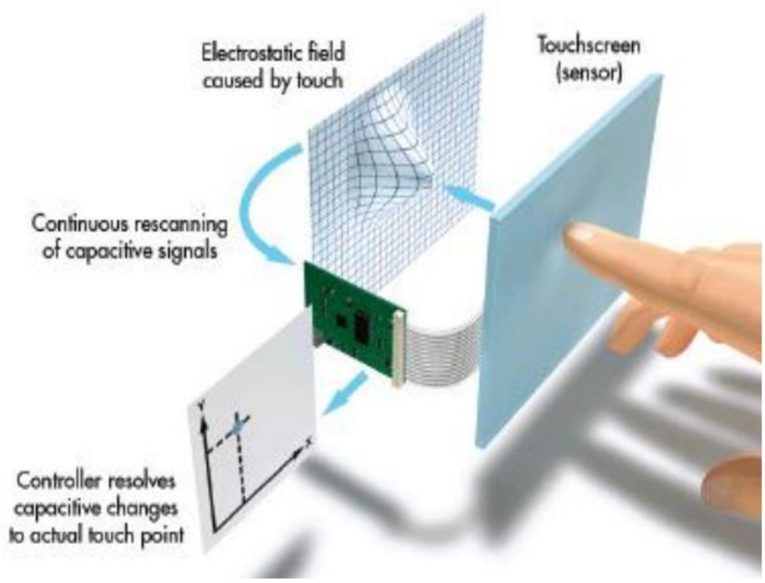

Figure 3.5.1

System Operation:

Our system has both hardware and software component. As such data collection is done in hardware and processing of data and subsequent controlling are done in the software and output circuit which are connected to electrical appliances.

Touchscreen:

There are two types of touchsceen currently available in market : capacitive touchscreen and resistive touchscreen. We are using a capacitive touchscreen for our project. Capacitive touchscreen rely on electrical properties of human body to detect where and when the user touches. Thus it has the advantage that capacitive displays can be controlled with very light finger touches and generally cannot be used with gloves or mechanical stylus. These touchscreen have been widely used for cell phone industry. A capacitive touchscreen has an insulator such as glass which is coated with indium tin oxide. Since a human body also acts as an electrical conductance, touching the screen results in a distortion of screen's electrostatic field, measured as a change in capacitance.

Software development:

The interfacing part is done with the help of pic 16f887. Since it has low price and wide range of application, high quality and is easily available, it is an ideal solution in applications such as: the control of different processes in industry, machine control devices, measurement of different values etc. Pic has internal a/d convertor which helps us to convert the analog coordinates from the touchscreen to digital so that it can be processed further. The x-y coordinates are supplied through 4 wires of touchscreen and the microcontroller process it to extract relevant information to determine whether a touch has occurred or not. This information helps us to determine the location of touch and make decision to send command to the output circuit.

Interfacing of output circuit:

For designing the output circuit we have employed relays. Relay is a switch that opens and closes under the influence of another electrical circuit. Buffers and encoders are also used to give us better driving capabilities. Along with touch we use infra red remote also. We control all the appliances with the help of touch and remote at the same time. In this project we use RC5 remote. PIC microcontroller provides an internal 10 bit adc and non voltatic memory to save the data . We use triac to vary the speed with zero cross detector. Triac is a power electronic component that can conduct in both directions when triggered through gate. On touch screen we mark some place for on/off and for speed control. For on off electrical appliances we use four space on touch screen. For AC load we use Triac as a main components.. For Triac control we use MOC 3021 driver IC. MOC 3021 provide a proper control signal from microcontroller. On touch screen we use increment and decrement button to increment and decrement the firing angle of the Triac. Whenever we use Triac driver for AC load we must need a zero cross over logic.The circuit diagram for the touchscreen control home automation is given below in Fig. 2. Output signal is available on he pin no 37,38,39,40 of the controller. On this pin we use LED as a output. On this LED output we use relay also.

4. HARDWARE AND SOFTWARE REQUIREMENT

4.1. Hardware

4.1.1. Arduino Mega 2560 Rev3:

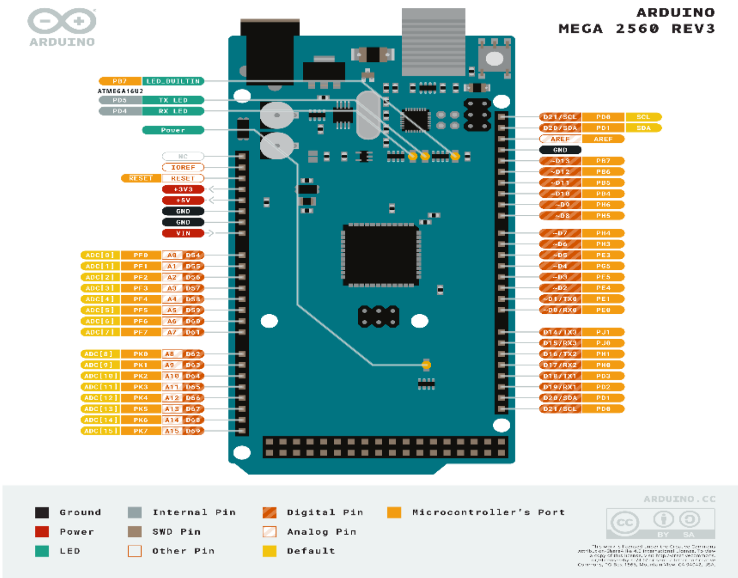

Figure 4.1.1

The Arduino Mega 2560 Rev3 is a powerful microcontroller board based on the ATmega2560. It is designed for complex projects that require more I/O pins, memory, and processing power than smaller boards like the Arduino Uno. Operating at 5V, it features 54 digital I/O pins (15 of which support PWM), 16 analog input pins, and multiple communication interfaces, including four UART serial ports, I2C, and SPI. With 256KB of Flash memory, 8KB of SRAM, and 4KB of EEPROM, it can handle large and advanced programs efficiently. The 16 MHz clock speed ensures smooth performance for real-time applications.

Powering the Arduino Mega 2560 is flexible, as it supports power input through a USB-B port, a DC power jack (7-12V recommended), or a Vin pin (7-12V). However, exceeding 12V is not advised as it may cause overheating. The board is widely used for robotics, home automation, industrial control, IoT applications, and data logging. Due to its large number of pins and memory, it is ideal for projects involving multiple sensors, motors, displays, and complex automation systems.

Programming the Arduino Mega 2560 is easy using the Arduino IDE, which supports numerous libraries and shields for expansion. It is compatible with most Arduino shields, allowing for easy integration with various modules like WiFi, Bluetooth, and motor controllers. A simple setup involves downloading the Arduino IDE, selecting the correct board and port, and writing/uploading code via the USB interface. A basic LED blink program can be written using digitalWrite() and delay() functions to test the board's functionality.

4.1.2. HC -05 Bluetooth Module:

Figure 4.1.2

The HC-05 is a widely used Bluetooth module designed for wireless communication between microcontrollers and other Bluetooth-enabled devices. Developed primarily for embedded systems, it enables short-range data transmission and is commonly utilized in various applications, including robotics, home automation, and IoT projects.

The HC-05 module operates in the 2.4 GHz frequency range and supports Bluetooth Serial Port Profile (SPP), allowing for seamless communication between devices like smartphones, tablets, and computers. It can function in both master and slave modes, giving developers the flexibility to establish communication in different configurations based on project requirements.

This module features a simple and intuitive serial interface, making it easy to connect to popular microcontrollers such as Arduino and Raspberry Pi. The HC-05 is equipped with a range of approximately 10 meters (33 feet) in open space, which is suitable for most short-range applications. Its low power consumption makes it ideal for battery-powered projects.

Programming the HC-05 is straightforward, typically requiring minimal configuration through AT commands for setting parameters such as baud rate, device name, and pairing settings. It is also compatible with a variety of platforms and development environments, further enhancing its versatility.

Due to its affordability, ease of integration, and widespread community support, the HC-05 Bluetooth module has become a popular choice among hobbyists and professionals alike.

Whether used for remote control applications, data logging, or sensor monitoring, the HC-05 serves as an effective solution for implementing wireless communication in embedded systems.

4.1.3. 4-Channel Relay Module:

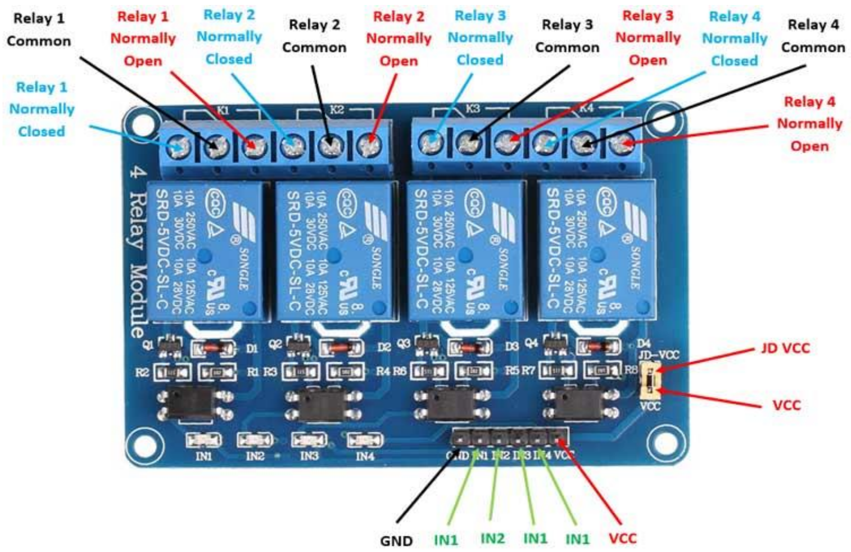

Figure 4.1.3

The 4-Channel Relay Module is a widely used component in various electronic projects, enabling the control of multiple high-voltage devices from low-voltage microcontrollers. It is particularly popular in home automation and industrial applications due to its ability to safely switch on/off appliances, lights, and motors.

The 4-Channel Relay Module is commonly used in home automation systems to control lighting, heating, and other appliances based on user-defined conditions or schedules. Its versatility and ease of use make it an ideal choice for implementing remote control solutions, automating machinery, and integrating smart appliances into IoT systems.

Features:

Each relay in the module acts as an electrically operated switch that can handle loads of up to 10A at 250VAC or 30VDC. This capability makes the 4-Channel Relay Module suitable for controlling a variety of devices without the need for complex wiring or circuitry. The module can be easily interfaced with popular microcontrollers like Arduino, Raspberry Pi, and ESP8266, making it accessible for both hobbyists and professionals.

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4.1.4. 5V, 2A Power Adapter:

Figure 4.1.4

The 5V, 2A Power Adapter is a compact and reliable power supply unit designed to convert standard AC voltage into a stable DC output suitable for powering various electronic devices and projects. It is commonly used in applications requiring a low voltage, making it ideal for powering microcontrollers, sensors, LED strips, and other low-power electronics.

This adapter provides a consistent output of 5 volts and a current capacity of 2 amps, making it suitable for devices that demand moderate power. The power adapter typically features a standard AC input range of 100-240V, allowing it to be used in different countries with varying voltage standards.

Features:

1. Voltage and Current Output:

- Delivers a stable 5V output with a maximum current of 2A, providing sufficient power for various low-power applications.

2. Universal Compatibility:

- Designed to work with a wide range of devices, including Arduino boards, Raspberry Pi, Wi-Fi modules, and other peripherals that require 5V power supply.

3. Safety Features:

- Equipped with built-in protections against over-voltage, over-current, and short circuits, ensuring safe operation and prolonged lifespan of connected devices.

4. Compact Design:

- Its small form factor makes it easy to transport and integrate into different setups without taking up much space.

4.1.5. TFT touch display:

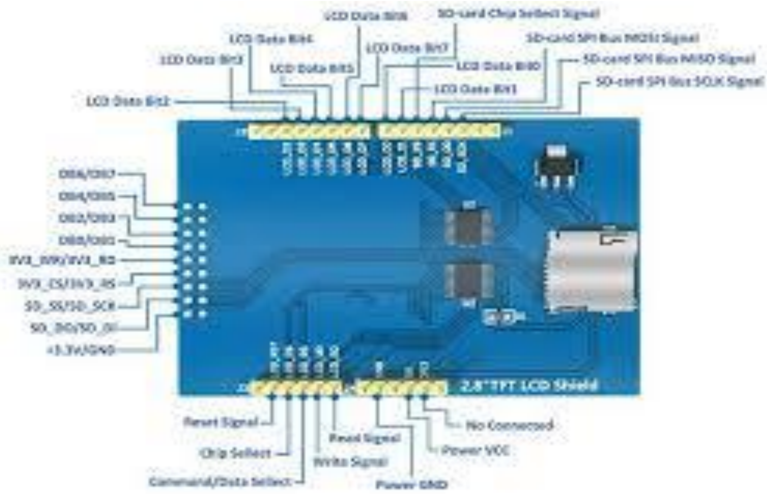

Figure 4.1.5

The TFT touch display is a compact and versatile electronic component designed to detect touch or proximity through human contact. It is widely used in various applications, including home automation, interactive projects, and user interface designs. This module typically consists of a capacitive or resistive touch sensor that can recognize a user's touch input and send a signal to a microcontroller or other devices, facilitating the creation of responsive electronic systems.

The touch sensor module is popular among hobbyists and developers due to its simplicity and ease of integration with microcontrollers like Arduino, Raspberry Pi, and ESP8266. By providing a touch-sensitive interface, it enables users to build interactive and user-friendly projects that enhance the overall experience of their applications.

Key Features:

1. Touch Sensitivity:

   The module is capable of detecting touch through skin contact or proximity, allowing for responsive interactions in various applications.

2. Digital Output:

   Typically provides a digital output signal (HIGH/LOW) that can be easily read by microcontrollers, enabling simple integration into projects.

3. Adjustable Sensitivity:

   Many touch sensor modules come with an adjustable sensitivity setting, allowing users to customize the touch detection threshold according to their requirements.

4. Compact Size:

Its small form factor allows for easy integration into tight spaces and various electronic projects, enhancing design flexibility.

5. Easy to Interface:

The module usually features straightforward pin connections, typically including power (VCC), ground (GND), and output (OUT), facilitating quick setup and integration with development boards.

4.1.6. IR Receiver:

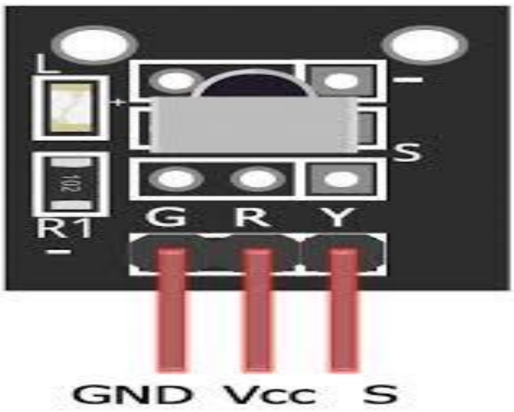

Figure 4.1.6

The IR Receiver is a sensor that detects infrared signals transmitted from remote controls or other IR sources, allowing for wireless communication with electronic devices. It is commonly used in applications such as remote control systems, home automation, and robotics.

4.1.7. IR Remote Control:

Figure 4.1.7

The IR Remote Control System enables users to wirelessly control various electronic devices, such as lights, fans, and appliances, using an infrared remote control. This project simplifies home automation by allowing convenient operation from a distance, enhancing user experience and accessibility.

4.1.8. Connecting Wires:

Figure 4.1.8

Connecting wires are essential electrical conductors used to establish connections between various components in electronic circuits. They come in different types, such as jumper wires, ribbon cables, and solid or stranded wire, and are crucial for transmitting power and signals between devices, ensuring proper functionality in projects like home automation, robotics, and prototyping. Their flexibility and versatility make them indispensable in both hobbyist and professional applications.

4.2 SOFTWARE:

4.2.1. ARDUINO BLUETOOTH CONTROL:

The Arduino Bluetooth Control app, allows users to wirelessly control Arduino-based projects using a smartphone via Bluetooth communication. It is commonly used with Bluetooth modules such as HC-05 and HC-06, which enable seamless serial data transmission between an Arduino board and an Android device. The app provides an intuitive interface that allows users to send commands to their Arduino projects in different ways, such as through buttons, voice commands, sliders, or terminal-based text inputs. This makes it a versatile tool for controlling a wide range of devices, including LEDs, motors, relays, smart home appliances, and robotics systems. Many versions of the app also feature real-time feedback, displaying sensor readings and system responses directly on the smartphone screen. Users can customize commands to suit specific project needs, making it highly adaptable for home automation, IoT applications, and DIY electronics projects. By eliminating the need for complex wiring, the Arduino Bluetooth Control app simplifies remote control functionality, offering a convenient and efficient way to interact with Arduino-based systems using just a smartphone.

4.2.2 ARDUINO IDE:

The Arduino IDE (Integrated Development Environment) is a free, open-source software used for writing, compiling, and uploading code to Arduino boards. Designed to be user-friendly, it supports programming in C and C++, making it accessible for both beginners and experienced developers. The IDE features a simple code editor with syntax highlighting, auto-indentation, and error detection, along with built-in libraries that provide support for various sensors, motors, and communication modules. It includes a compiler that converts written code into machine language, allowing seamless communication between the computer and the microcontroller. The Serial Monitor and Serial Plotter functions enable real-time debugging and data visualization, making it easier to troubleshoot projects.

Arduino IDE is compatible with Windows, macOS, and Linux and supports multiple Arduino boards, including the Uno, Mega, and Nano, as well as third-party microcontrollers. Users can program an Arduino board by writing code, verifying it for errors, compiling it, and uploading it through a USB connection. The IDE provides an easy way to interact with hardware, making it suitable for applications such as home automation, robotics, IoT, and embedded systems. With both offline and online versions available, users can choose between the standalone desktop

IDE or the Arduino Web Editor for cloud-based development. The latest versions, including Arduino IDE 2.0, introduce improvements such as a modern interface, enhanced debugging tools, and faster compilation speeds. Being open-source and community-driven, the Arduino IDE continues to evolve, providing a powerful yet straightforward platform for developing microcontroller-based projects.

Arduino Mega or Mega 2...

sketch_mar27a.ino

```

#define RELAY_1_ON  0xFC03EF00
#define RELAY_1_OFF 0xFD02EF00
#define RELAY_2_ON  0xFB04EF00
#define RELAY_2_OFF 0xFA05EF00
#define RELAY_3_ON  0xF906EF00
#define RELAY_3_OFF 0xF807EF00
#define RELAY_4_ON  0xF708EF00
#define RELAY_4_OFF 0xF609EF00

void drawButtons() {
  tft.fillScreen(BLACK);
  for (int i = 0; i < 4; i++) {
    tft.fillRect(btnX[i], btnY[i], BTN_W, BTN_H, RED);  // Always red
    tft.setTextColor(WHITE);
    tft.setTextSize(2);

    // Center text
    int textX = btnX[i] + 20;
    int textY = btnY[i] + 20;
    tft.setCursor(textX, textY);
    tft.print(labels[i][0]);

    if (strlen(labels[i][1]) > 0) {
      tft.setCursor(textX, textY + 20);
      tft.print(labels[i][1]);
    }
  }
}

String readvoice;

void setup() {
  Serial.begin(9600);
  BT_SERIAL.begin(9600);
  IrReceiver.begin(IR_RECEIVE_PIN, ENABLE_LED_FEEDBACK);

  uint16_t ID = tft.readID();
  tft.begin(ID);
  tft.setRotation(1);
  SCREEN_W = tft.width();
  SCREEN_H = tft.height();

  for (int i = 0; i < 4; i++) {
```

Arduino Mega or Mega 2...

sketch_mar27a.ino

```
  for (int i = 0; i < 4; i++) {
    pinMode(relayPins[i], OUTPUT);
    digitalWrite(relayPins[i], HIGH);
  }

  drawButtons();
}

void loop() {
  TSPoint p = ts.getPoint();

  if (p.z > 300) {  // Increase touch detection threshold
    pinMode(XM, OUTPUT);
    pinMode(YP, OUTPUT);
    int touchX = map(p.y, 150, 900, 0, SCREEN_W);
    int touchY = map(p.x, 150, 900, 0, SCREEN_H);

    Serial.print("Touch X: "); Serial.print(touchX);
    Serial.print(" Y: "); Serial.println(touchY);

    for (int i = 0; i < 4; i++) {
      if ((touchX >= btnX[i]) && (touchX <= btnX[i] + BTN_W) &&
          (touchY >= btnY[i]) && (touchY <= btnY[i] + BTN_H)) {

        delay(300);  // Debounce
        relayState[i] = !relayState[i];
        digitalWrite(relayPins[i], relayState[i] ? HIGH : LOW);
        break;
      }
    }
  }

  while (BT_SERIAL.available()) {
    delay(3);
    char c = BT_SERIAL.read();
    readvoice += c;
  }

  if (readvoice.length() > 0) {
    Serial.print("Received Bluetooth: "); Serial.println(readvoice);
    if (readvoice == "blue light on") digitalWrite(A8, LOW);
    if (readvoice == "blue light off") digitalWrite(A8, HIGH);
    if (readvoice == "fan on") digitalWrite(A10, LOW);
    if (readvoice == "fan off") digitalWrite(A10, HIGH);
```

```
    }

    if (readvoice.length() > 0) {
        Serial.print("Received Bluetooth: "); Serial.println(readvoice);
        if (readvoice == "blue light on") digitalWrite(A8, LOW);
        if (readvoice == "blue light off") digitalWrite(A8, HIGH);
        if (readvoice == "fan on") digitalWrite(A10, LOW);
        if (readvoice == "fan off") digitalWrite(A10, HIGH);
        if (readvoice == "green light on") digitalWrite(A11, LOW);
        if (readvoice == "green light off") digitalWrite(A11, HIGH);
        if (readvoice == "switch on") digitalWrite(A9, LOW);
        if (readvoice == "switch off") digitalWrite(A9, HIGH);
        readvoice = "";
    }

    if (IrReceiver.decode()) {
        uint32_t receivedCode = IrReceiver.decodedIRData.decodedRawData;
        Serial.print("Received IR Code: ");
        Serial.println(receivedCode, HEX);
        controlRelay(receivedCode);
        IrReceiver.resume();
    }
}

void controlRelay(uint32_t command) {
    switch (command) {
        case RELAY_1_ON: digitalWrite(A8, LOW); Serial.println("Relay 1 ON"); break;
        case RELAY_1_OFF: digitalWrite(A8, HIGH); Serial.println("Relay 1 OFF"); break;
        case RELAY_2_ON: digitalWrite(A10, LOW); Serial.println("Relay 2 ON"); break;
        case RELAY_2_OFF: digitalWrite(A10, HIGH); Serial.println("Relay 2 OFF"); break;
        case RELAY_3_ON: digitalWrite(A11, LOW); Serial.println("Relay 3 ON"); break;
        case RELAY_3_OFF: digitalWrite(A11, HIGH); Serial.println("Relay 3 OFF"); break;
        case RELAY_4_ON: digitalWrite(A9, LOW); Serial.println("Relay 4 ON"); break;
        case RELAY_4_OFF: digitalWrite(A9, HIGH); Serial.println("Relay 4 OFF"); break;
        default: Serial.println("Unknown Command"); break;
    }
}
```

6.CIRCUIT DIAGRAM AND DESCRIPTION

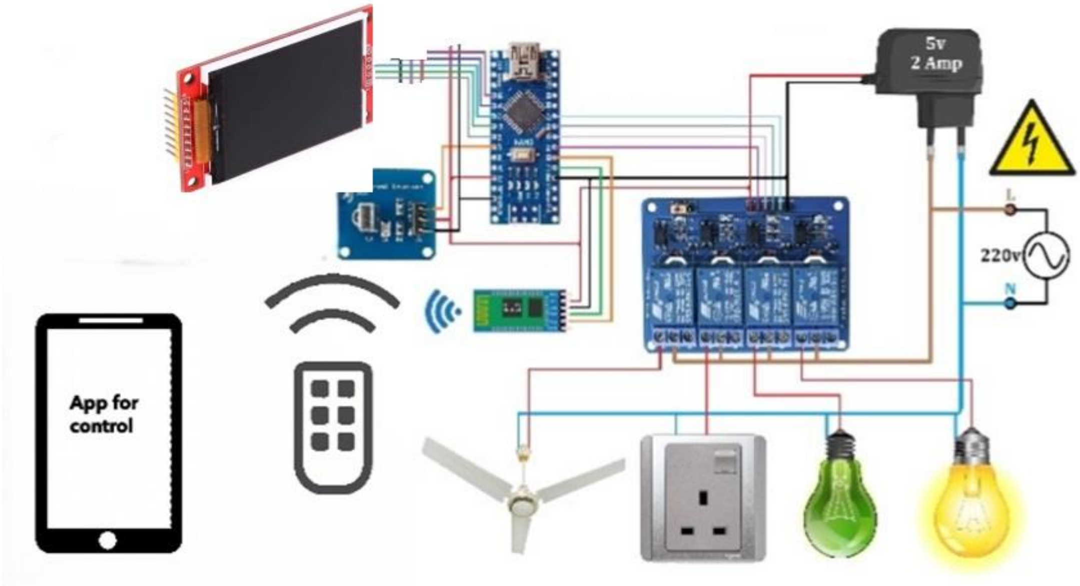

Figure. 6.1

Components used in the experimental setup as follows.

The components in the system shown in the image are part of a "Wireless Touch Screen Based Automation System" involving home appliance control via a smartphone app. Based on the diagram, here is a list of components:

1. Arduino Mega: The central processing unit for wireless communication and control.

2. HC-05 Bluetooth Module: For wireless communication between the smartphone app and Arduino.

3. 4-Channel Relay Module: To control high-voltage appliances (light bulbs, fan, etc.) by switching them on/off.

4. 5V, 2A Power Adapter: To supply power to the system, particularly to the Arduino and relay module.

5. 220V AC Power Source: For powering the household appliances.

6. Touchscreen Display: Connected to Arduino for user interaction (used to control the system directly from the display).

7. Appliances:

- Light Bulb (2): To represent different load outputs controlled by the relay.
- Fan: Another appliance connected for control.
- Power Socket: Represents a general power outlet that can also be controlled.

8. Android Smartphone with App: For controlling the appliances via Bluetooth.

9. Remote Control: Receives signal from IR remote and Sends commands to control appliances

10. Power Distribution Wires and Grounding: Various connections between the components for power and control.

Each of these components plays a role in creating a fully automated, wireless system for home appliance control via Bluetooth and a touch interface.

6. PROCEDURE AND WORKING

6.1. Working:

6.1.1. Powering the system

-The entire system is powered by a 5V, 2A adapter that supplies voltage to the Arduino Mega and the relay module.

-The electrical appliances (like the fan, lights, and socket) connected to the 220V AC mains are controlled through the relay module.

6.1.2. Bluetooth App-Based Control:

-The Android app is connected to the system via Bluetooth.

-When the user presses a control button in the app (e.g., ON/OFF for light, fan, socket), the command is transmitted wirelessly to the Arduino Mega through the HC-05 Bluetooth module.

-The Arduino processes the command and toggles the corresponding relay to switch the appliance ON or OFF.

6.1.3. Manual Control via Touch Sensors:

-Touch sensors are connected to the Arduino, each representing an individual appliance (fan, light, socket).

-When the user touches a sensor, a signal is sent to the Arduino.

-The Arduino then activates the respective relay to turn the appliance ON or OFF.

6.1.4. IR Remote Control:

-The system is also operable through an IR remote. When a button on the remote is pressed, the IR receiver sends the corresponding signal to the Arduino.

-The Arduino decodes the signal and triggers the relevant relay to switch the appliance ON or OFF.

6.1.5. Relay Switching:

-The Arduino controls the relay module by sending a HIGH or LOW signal to each relay pin.

-A HIGH signal activates the relay, closing the circuit and allowing 220V AC to power the connected appliance (light, fan, socket).

-A LOW signal deactivates the relay, breaking the circuit and turning off the appliance.

6.1.6. Appliance Control:

-The system controls different appliances like fans, lights, and sockets, either via the app, touch sensors, or IR remote.

-Once the Arduino receives a command (from any of the input methods), it checks which appliance is to be controlled and sends a control signal to the corresponding relay. The appliance either turns ON or OFF based on the current state.

6.1.7. Voice Command:

-The app also has an option for voice commands. When the user gives a voice command (e.g., "Turn ON light"), the app sends this command via Bluetooth to the Arduino.

-The Arduino then processes the voice command and toggles the appropriate relay to control the appliance.

6.2. Flow of Operation:

6.2.1. User Interaction:

The user can interact with the system through three methods: mobile app, touch sensors, or IR remote.

6.2.2. Command Processing:

Each method sends a signal to the Arduino Mega, which acts as the main processing unit.

6.2.3. Appliance Control:

Based on the input, the Arduino toggles the correct relay to control the connected appliance (fan, light, socket).

7.RESULT AND DISCUSSION

7.1. **Touch Sensor Responsiveness**:

The touch sensors should be highly responsive, registering touch inputs promptly and toggling the respective relays to control the appliances. Users should be able to switch appliances ON or OFF by a single touch without noticeable delay or failure.

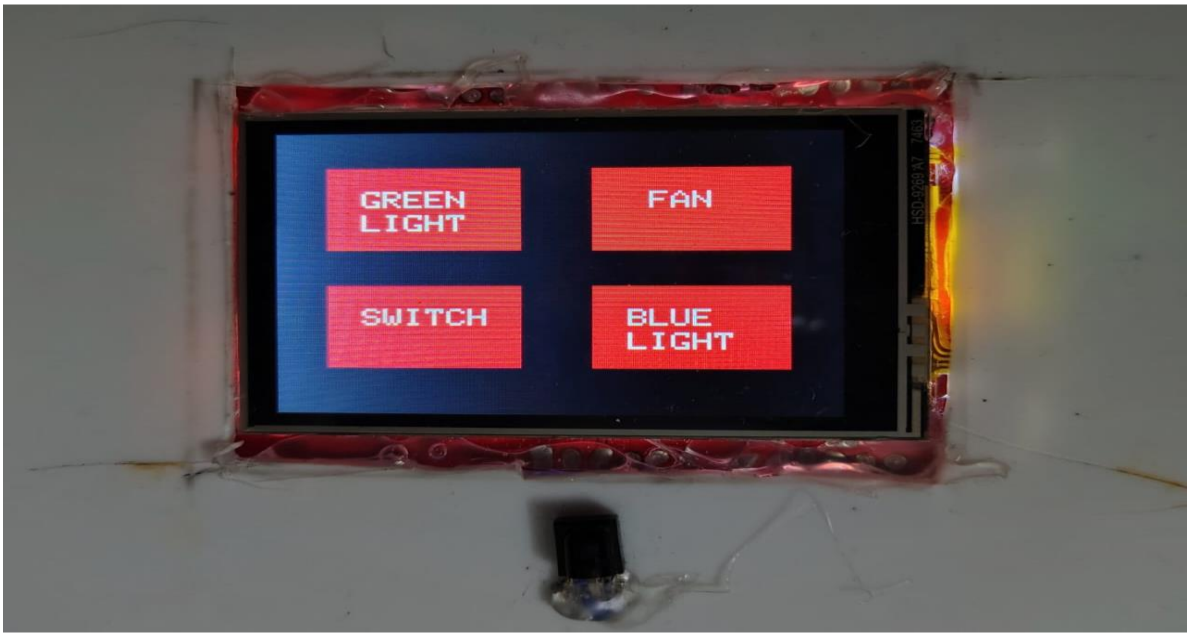

Figure 7.1.1

Figure 7.1.2

7.2. IR Remote Control:

The IR receiver should be able to decode signals from an IR remote accurately. The system should respond to remote inputs from within typical line-of-sight distances (around 5-10 meters), allowing the user to control the appliances without needing to use the app or touch sensors.

Figure 7.2.1

Figure 7.2.2

7.3. **Bluetooth Communication**:

The **Arduino Bluetooth Control** mobile app should successfully connect to the system via the **HC-05 Bluetooth module**. The Bluetooth communication is expected to work reliably within a range of approximately 10 meters, allowing wireless control of the devices within this distance.

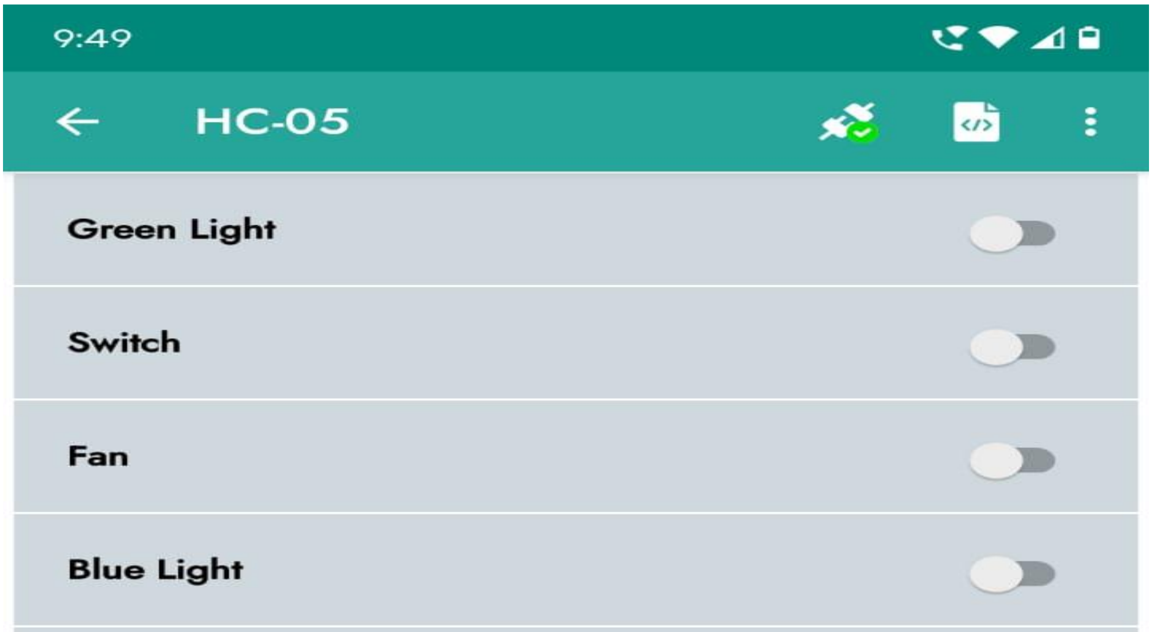

Figure 7.3.1

Figure 7.3.2

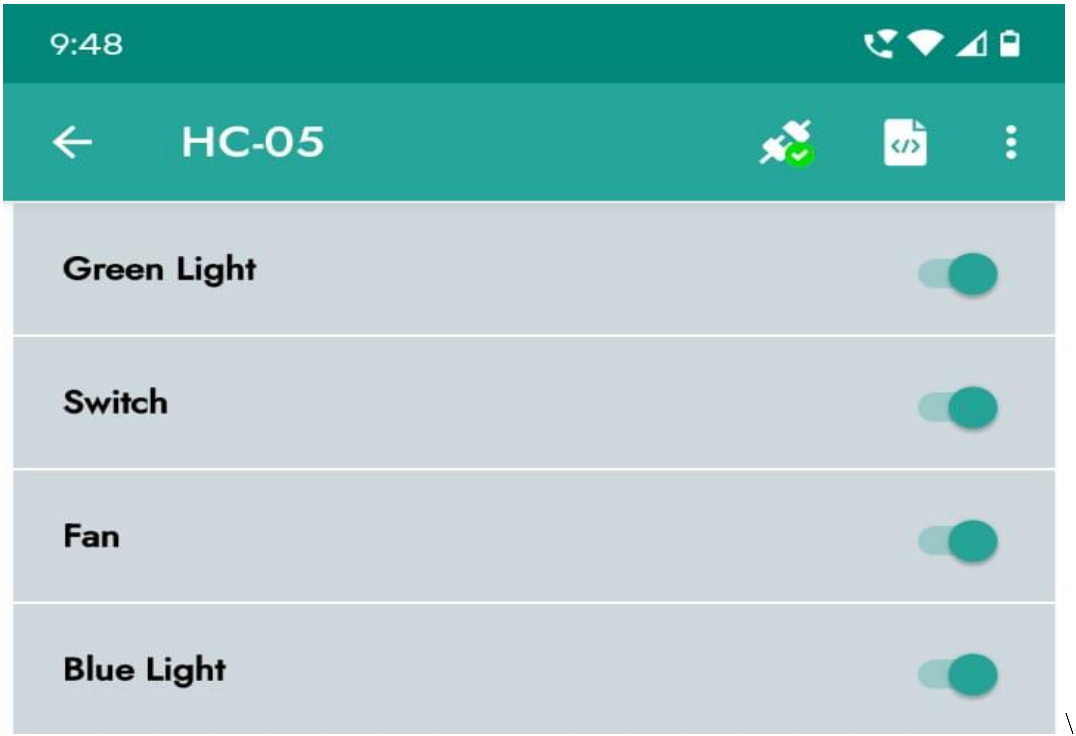

Figure 7.3.3

Figure 7.3.4

7.4. Voice Command:

The app also has an option for voice commands. When the user gives a voice command (e.g., "Turn ON light"), the app sends this command via Bluetooth to the Arduino.

The Arduino then processes the voice command and toggles the appropriate relay to control the appliance.

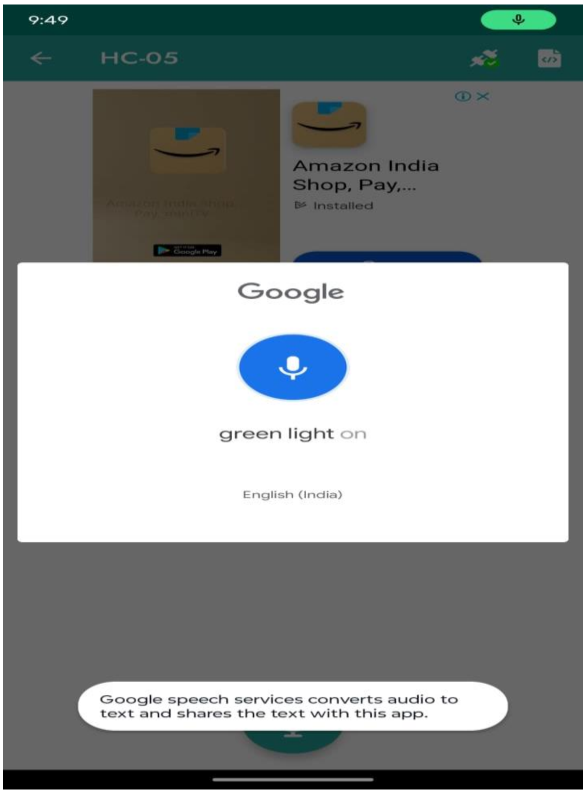

Figure 7.4.1

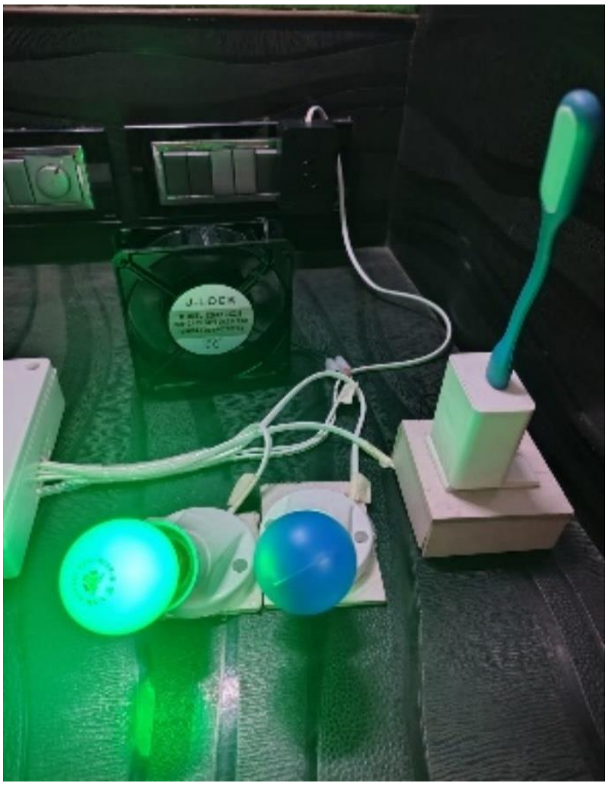

Figure 7.4.2

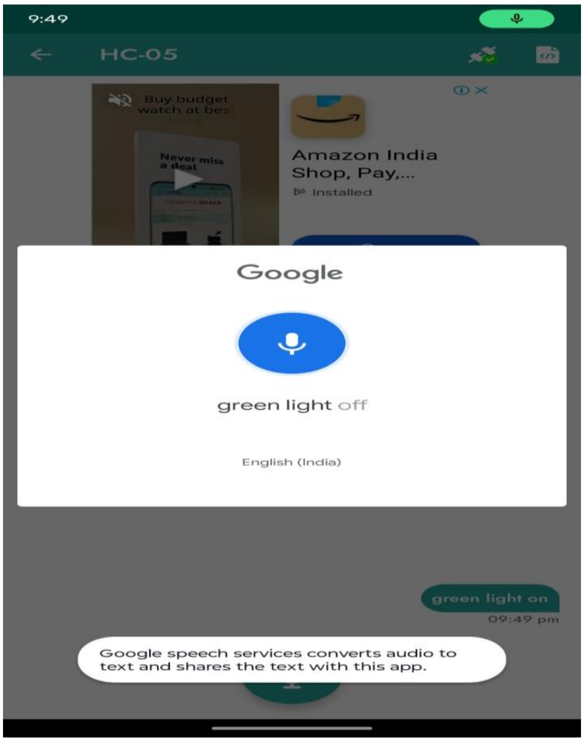

Figure 7.4.3

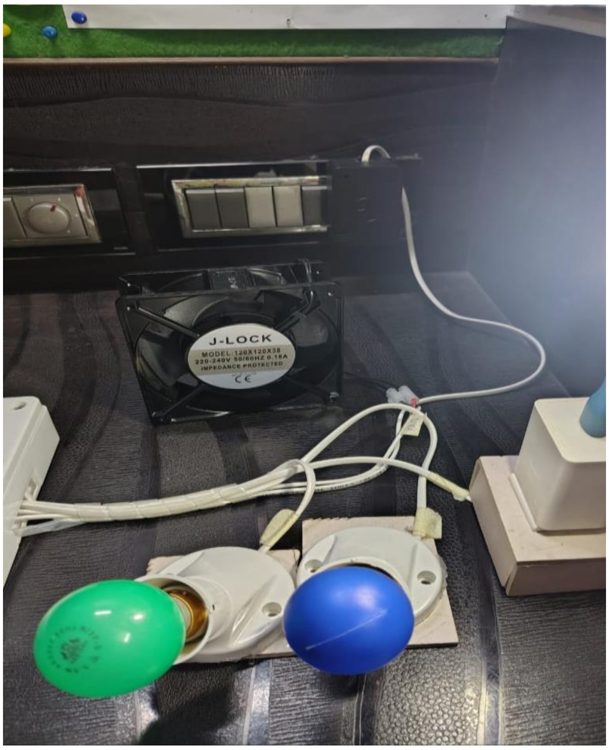

Figure 7.4.4

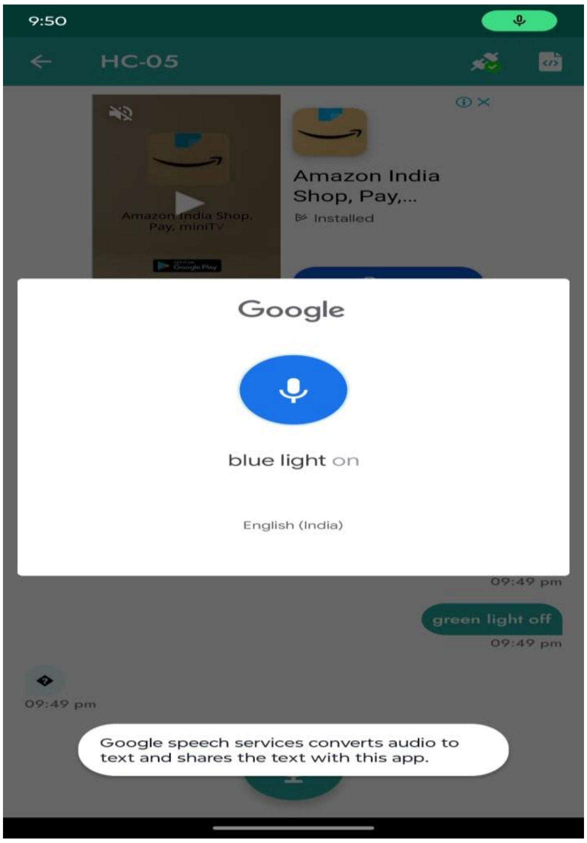

Figure 7.4.5

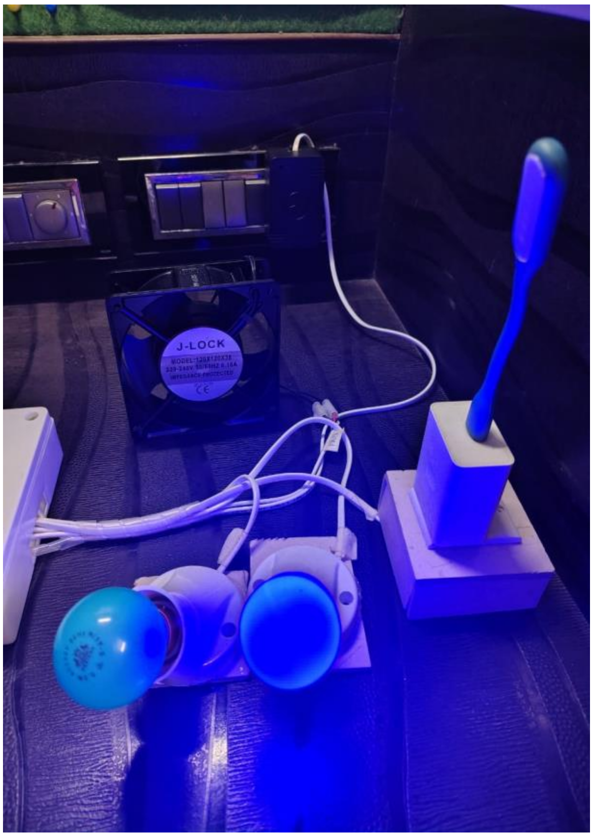

Figure 7.4.6

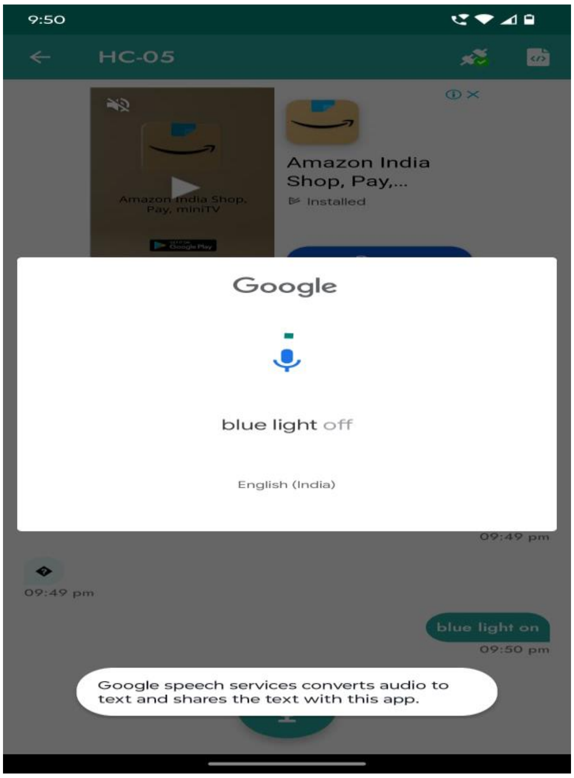

Figure 7.4.7

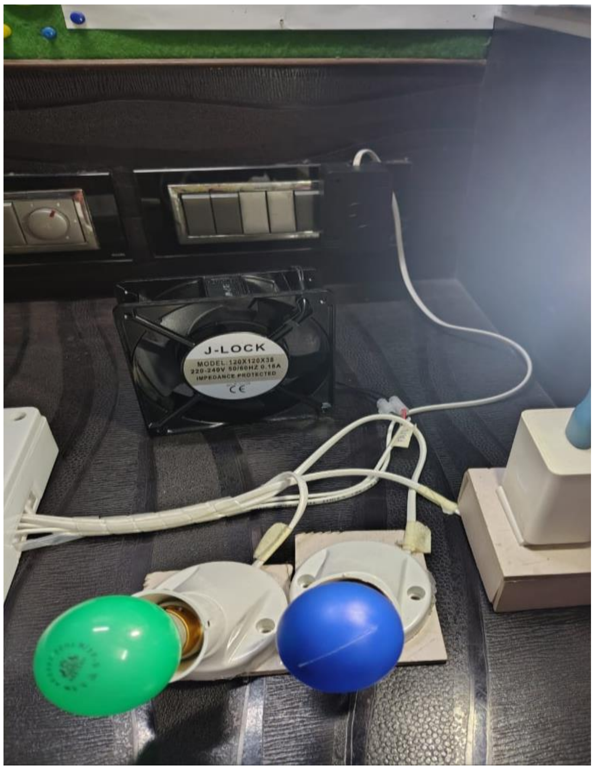

Figure 7.4.8

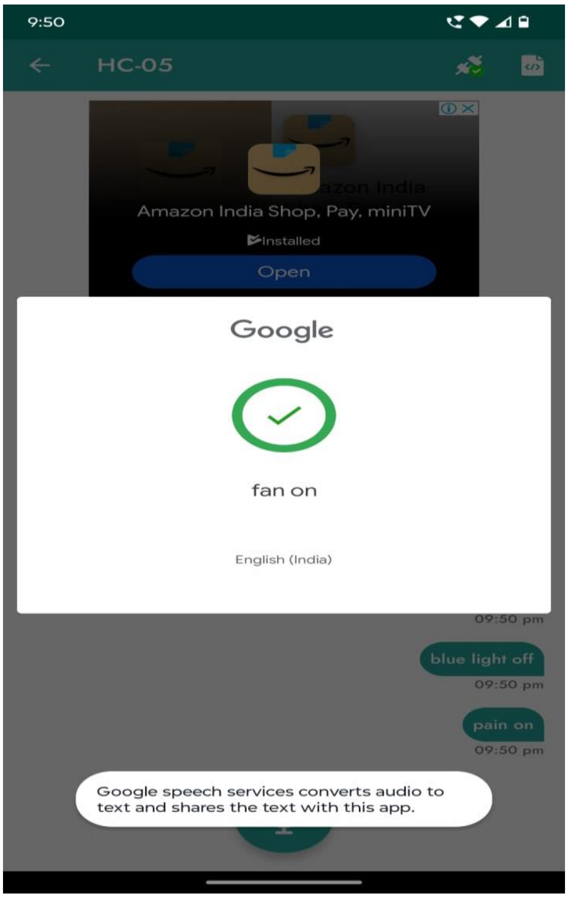

Figure 7.4.9

Figure 7.4.10

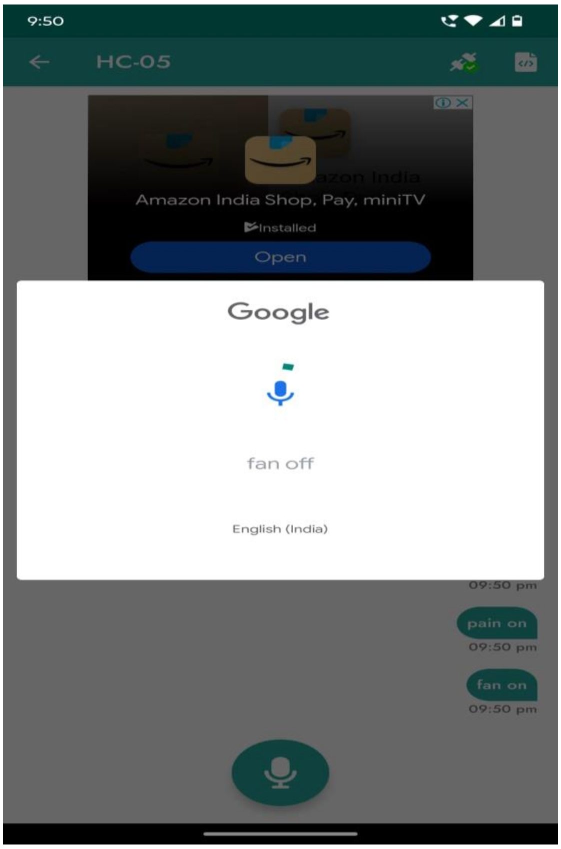

Figure 7.4.11

Figure 7.4.12

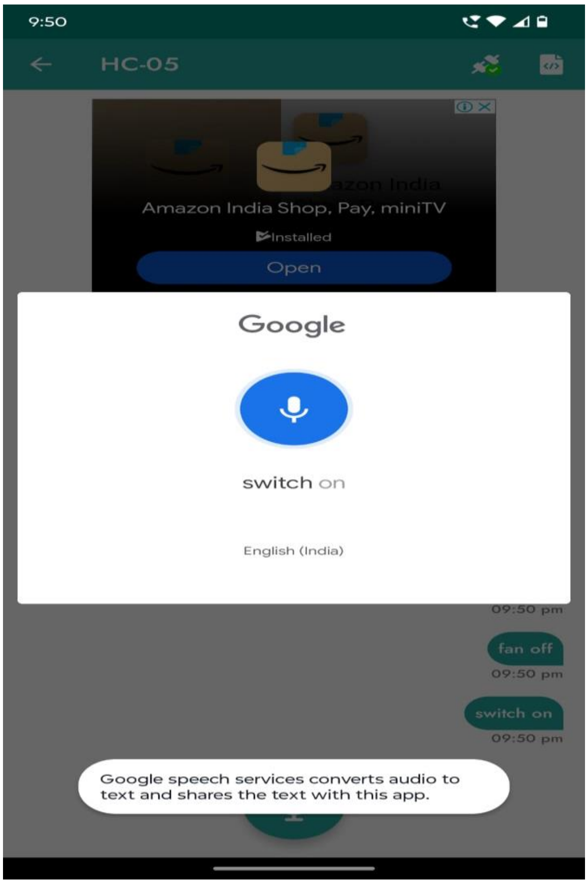

Figure 7.4.13

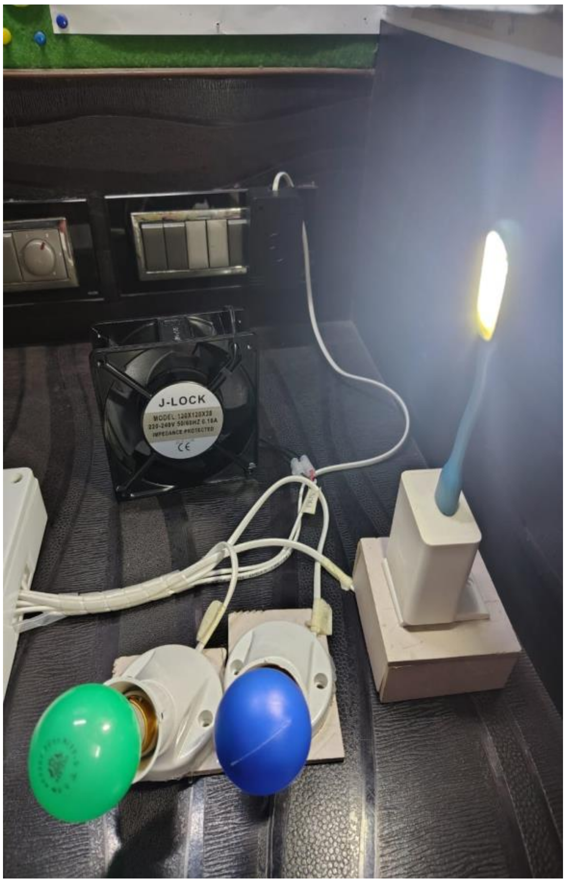

Figure 7.4.14

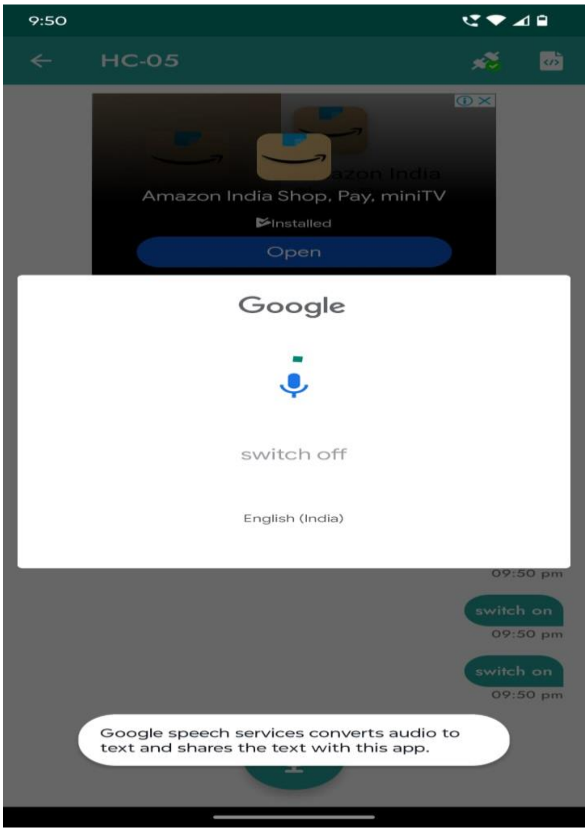

Figure 7.4.15

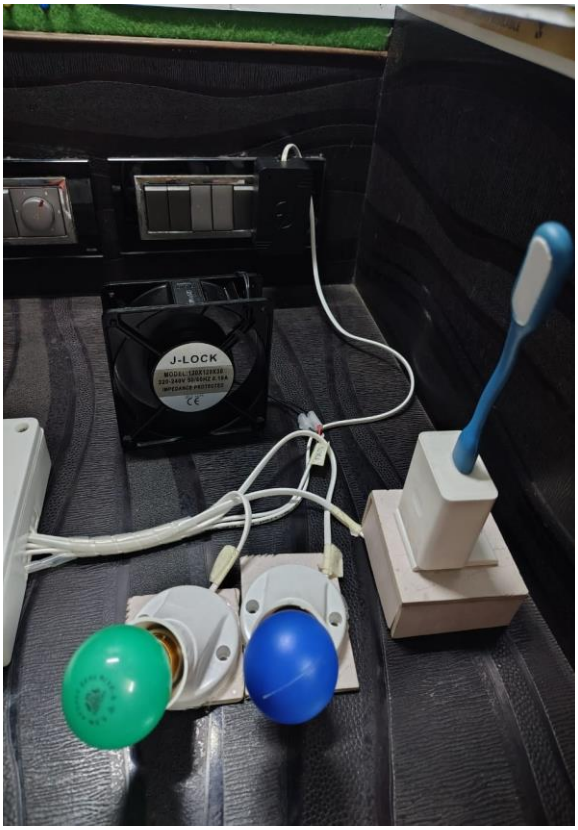

Figure 7.4.16

8. CONCLUSION

The Arduino Mega stands out as a powerful and compact microcontroller board, making it an excellent choice for a wide variety of electronics projects, especially where space and efficiency are critical. Its compatibility with numerous sensors, modules, and communication devices allows for seamless integration in home automation, IoT, and robotics applications. With its easy programmability via the Arduino IDE and widespread community support, the Arduino Nano is both beginner-friendly and versatile, enabling users to innovate and create efficient, reliable systems. Whether for hobbyists or professionals, it offers a perfect balance between functionality and simplicity, making it a go-to solution for embedded system projects.

8.1. **Reliable Control of Appliances**:
The system should allow seamless control of connected appliances (light, fan, socket, etc.) using the three methods—Bluetooth mobile app, touch sensors, and IR remote. The appliances should respond immediately to ON/OFF commands with minimal delay.

8.2. **Efficient Power Management**:
The system should operate stably with a 5V, 2A power supply, providing sufficient power for the Arduino Mega, relay module, and other components. There should be no unexpected resets or shutdowns due to power issues.

8.3. **Simultaneous Control**:
The system should support simultaneous use of different control methods (e.g., using the touch sensor while also having the Bluetooth app connected). The Arduino should handle inputs from all three control interfaces without conflicts or delays.

8.4. **Accurate Relay Operation**:
The 4-channel relay module should switch appliances ON and OFF as directed by the Arduino Mega. The relays should safely handle the 220V AC load and provide isolation between the control and high-voltage sections, ensuring safe operation.

8.5. **User-Friendly Interface**:
The mobile app interface should be simple and intuitive, allowing users to easily control the appliances with ON/OFF buttons for each device, and the voice command feature should work smoothly.

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